# VTPEH 6270 - Check Point 06

Statistical Analyses

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# 1 Introduction

## 1.1 Research Question

This study examines whether the prevalence of overweight classification differed by race/ethnicity among children participating in WIC in the United States in 2008.

## 1.2 Scientific Plausibility and Context

Childhood overweight and obesity remain important public health concerns in the United States because excess weight in childhood is associated with important health consequences, and national surveillance data show that obesity affects a substantial share of U.S. children. Racial and ethnic differences in obesity prevalence have also been documented at the national level, suggesting that weight-related outcomes are not evenly distributed across populations (Centers for Disease Control and Prevention 2024). Prior research further indicates that racial and ethnic disparities in obesity are already present in early childhood, supporting the scientific relevance of examining these patterns before adolescence (Isong et al. 2018). This issue is especially relevant in low-income populations, because studies of low-income preschool-aged children have found persistent racial and ethnic differences in obesity prevalence across U.S. states and territories (Pan et al. 2016). Therefore, it is scientifically plausible to examine whether the prevalence of overweight classification differed by race/ethnicity among children participating in WIC in the United States in 2008.

# 2 Materials and Methods

## 2.1 Data

The data used in this analysis were obtained from Data.CDC.gov. Specifically, I used the dataset *Nutrition, Physical Activity, and Obesity - Women, Infant, and Child*, which contains information on weight status for children aged 3 months to 4 years from Women, Infant, and Child Participant and Program Characteristics (WIC-PC). According to the dataset description, these data are used in CDC's Data, Trends, and Maps database to provide national- and state-level information related to obesity, nutrition, and physical activity.

For this study, I focused on records from year of 2008 and examined whether the prevalence of overweight classification differed by race/ethnicity among children participating in WIC in the United States. Relevant variables were retained for year, race/ethnicity, and overweight prevalence. Additional data cleaning and subsetting steps are described below.

```r
# packages
library(readr)
library(ggplot2)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
#load data
setwd("/Users/eve/Desktop/V270")
wic <- read_csv("888/checkpoint 6/Nutrition,_Physical_Activity,_and_Obesity_-_Women,_Infant,_and_Child_
```

```
## Rows: 12852 Columns: 31

## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (23): LocationAbbr, LocationDesc, Datasource, Class, Topic, Question, Da...
## dbl  (6): YearStart, YearEnd, Data_Value, Data_Value_Alt, Low_Confidence_Lim...
## num  (1): Sample_Size
## lgl  (1): Data_Value_Unit
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
# clean data
wic_2008 <- subset(wic, YearStart == 2008)
wic_2008 <- subset(wic_2008, Question == "Percent of WIC toddlers who have an overweight classification
wic_2008 <- subset(wic_2008, StratificationCategory1 == "Race/Ethnicity")
```

```r
head(wic_2008)

wic_2008_clean <- subset(wic_2008, !is.na(Data_Value))

wic_2008_clean <- wic_2008_clean[, c(
  "YearStart",
  "LocationDesc",
  "Question",
  "StratificationCategory1",
  "Stratification1",
  "Data_Value"
)]
```

### 2.2 Statistical Analyses

To examine whether overweight prevalence differed by race/ethnicity among WIC toddlers in the United States in 2008, I treated the prevalence estimate (`Data_Value`) as the outcome variable and race/ethnicity as the explanatory grouping variable. Because the analysis compared mean prevalence estimates across more than two racial/ethnic groups, a one-way analysis of variance (ANOVA) was considered as the primary statistical approach.

Before conducting ANOVA, I assessed whether the data reasonably met the assumptions of the test, including approximate normality of residuals and homogeneity of variance across groups. If these assumptions were not adequately met, I planned to use the Kruskal-Wallis test as a nonparametric alternative. If the overall group comparison was statistically significant, post-hoc pairwise comparisons were conducted to identify which racial/ethnic groups differed from one another. Statistical significance was evaluated at an alpha level of 0.05.

### 2.3 Code

All related files are in this link: https://github.com/yiz228/888 This repository includes the dataset, R markdown PDF report and script, the bibliography and README.

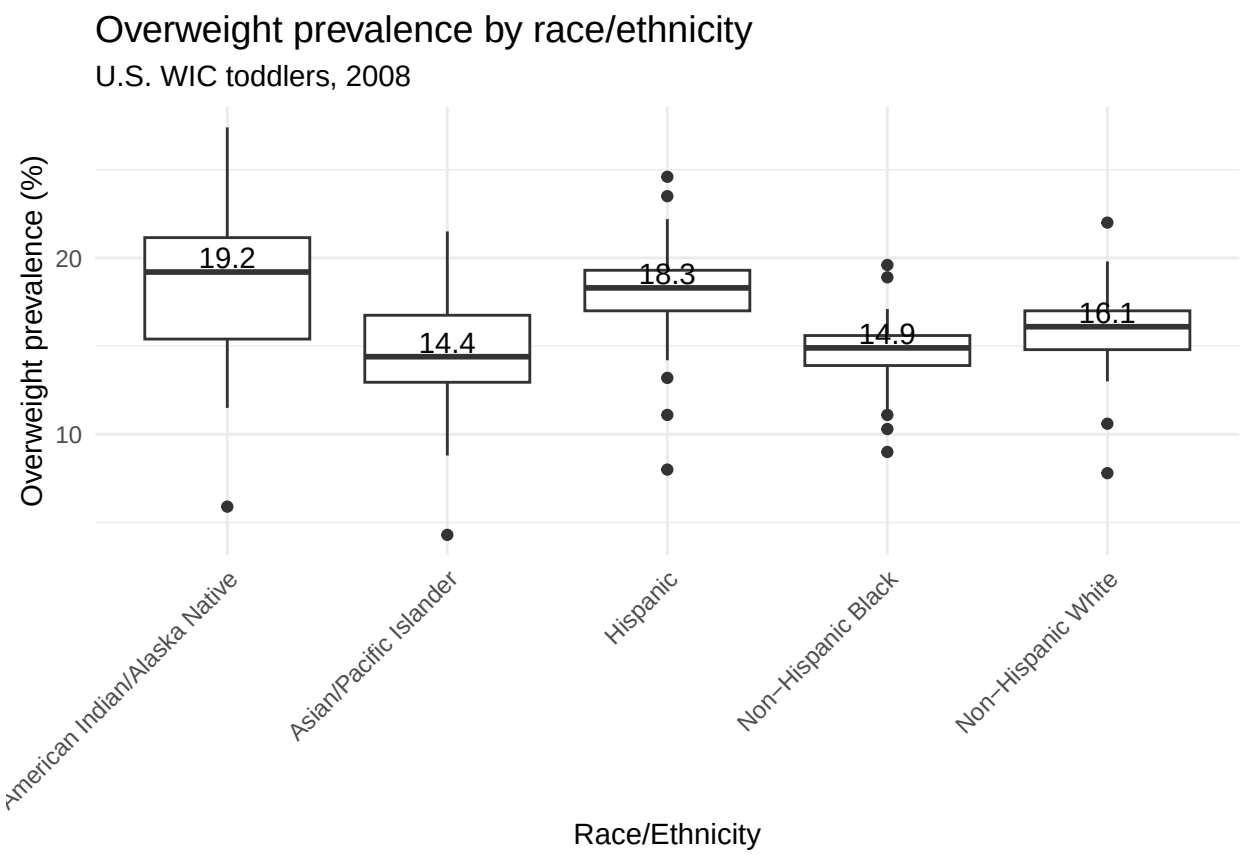
## 3 Results

### 3.1 Visualization

```r
median_df <- wic_2008_clean %>%
  group_by(Stratification1) %>%
  summarise(median_value = median(Data_Value, na.rm = TRUE))

ggplot(wic_2008_clean, aes(x = Stratification1, y = Data_Value)) +
  geom_boxplot() +
 geom_text(
  data = median_df,
  aes(x = Stratification1, y = median_value + 0.8, label = round(median_value, 1)),
  size = 4
) +
  labs(
```

```
    title = "Overweight prevalence by race/ethnicity",
    subtitle = "U.S. WIC toddlers, 2008",
    x = "Race/Ethnicity",
    y = "Overweight prevalence (%)"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1, size = 9),
    axis.title.x = element_text(size = 11),
    axis.title.y = element_text(size = 11),
    plot.title = element_text(size = 14),
    plot.subtitle = element_text(size = 11)
  )
```




The boxplot suggests that overweight prevalence differed across racial/ethnic groups among U.S. WIC toddlers in 2008. Median prevalence appeared highest among American Indian/Alaska Native toddlers (19.2%) and Hispanic toddlers (18.3%), and lowest among Asian/Pacific Islander toddlers (14.4%). Non-Hispanic Black toddlers had a median prevalence of 14.9%, while Non-Hispanic White toddlers had a median prevalence of 16.1%. Overall, the visualization suggests meaningful variation in overweight prevalence across racial/ethnic groups.

### 3.2 Statistical Results

```r
kruskal.test(Data_Value ~ Stratification1, data = wic_2008_clean)
```

```
##
##  Kruskal-Wallis rank sum test
##
## data:  Data_Value by Stratification1
## Kruskal-Wallis chi-squared = 70.412, df = 4, p-value = 1.858e-14
```

```r
pairwise.wilcox.test(
  wic_2008_clean$Data_Value,
  wic_2008_clean$Stratification1,
  p.adjust.method = "bonferroni"
)
```

```
##
##  Pairwise comparisons using Wilcoxon rank sum test with continuity correction
##
## data:  wic_2008_clean$Data_Value and wic_2008_clean$Stratification1
##
##                        American Indian/Alaska Native Asian/Pacific Islander
## Asian/Pacific Islander 1.4e-05                       -
## Hispanic               1.0000                        5.7e-07
## Non-Hispanic Black     7.1e-06                       1.0000
## Non-Hispanic White     0.0025                        0.0421
##                        Hispanic Non-Hispanic Black
## Asian/Pacific Islander -        -
## Hispanic               -        -
## Non-Hispanic Black     9.5e-10  -
## Non-Hispanic White     4.1e-06  0.0020
##
## P value adjustment method: bonferroni
```

A Kruskal-Wallis rank sum test was conducted to compare overweight prevalence across racial/ethnic groups among U.S. WIC toddlers in 2008. The test showed a statistically significant difference in overweight prevalence across groups, x-squared (4) = 70.412, $p < 0.001$. This result suggests that overweight prevalence differed across racial/ethnic groups among U.S. WIC toddlers in 2008.

Post-hoc pairwise Wilcoxon rank sum tests with Bonferroni adjustment indicated several significant differences between groups. American Indian/Alaska Native toddlers had significantly different overweight prevalence from Asian/Pacific Islander toddlers ($p < 0.001$), Non-Hispanic Black toddlers ($p < 0.001$), and Non-Hispanic White toddlers ($p = 0.0025$), but not from Hispanic toddlers ($p = 1.000$). Hispanic toddlers also differed significantly from Asian/Pacific Islander toddlers ($p < 0.001$), Non-Hispanic Black toddlers ($p < 0.001$), and Non-Hispanic White toddlers ($p = 0.0020$). In contrast, Asian/Pacific Islander and Non-Hispanic Black toddlers did not differ significantly ($p = 1.000$).

# 4 Conclusion

The results of this analysis indicate that overweight prevalence differed significantly by race/ethnicity among U.S. WIC toddlers in 2008. Overweight prevalence appeared highest among American Indian/Alaska Native and Hispanic toddlers and lowest among Asian/Pacific Islander toddlers, with Non-Hispanic Black and Non-Hispanic White toddlers falling in between. These findings are consistent with the expectation stated in the introduction that weight-related outcomes may not be distributed equally across racial and ethnic groups.

More broadly, the results align with prior literature suggesting that disparities in childhood overweight and obesity emerge early in life and are present across racial and ethnic groups (Centers for Disease Control and Prevention 2024; Isong et al. 2018; Pan et al. 2016). In this sense, the findings support the scientific plausibility of examining racial/ethnic differences in overweight prevalence within the WIC population. At the same time, this analysis was limited to one year of state-level prevalence estimates and therefore cannot explain the causes of these differences or make conclusions about individual-level risk. Future analyses could examine patterns across multiple years or explore other factors associated with overweight prevalence in this population.

Overall, this analysis suggests that overweight prevalence among U.S. WIC toddlers in 2008 differed meaningfully across racial/ethnic groups, highlighting the importance of continued attention to disparities in early childhood health.

# 5 AI Use Disclosure Statement

ChatGPT was used to help troubleshoot reported errors, assist with generating and revising R code, and provide guidance on organizing files and uploading materials to GitHub.

# References


Centers for Disease Control and Prevention. 2024. "Childhood Obesity Facts." https://www.cdc.gov/obesity/childhood-obesity-facts/childhood-obesity-facts.html.

Isong, Inyang A., Srinath R. Rao, Marie-Helene C. Bind, Mauricio Avendaño, Ichiro Kawachi, and Tracy K. Richmond. 2018. "Racial and Ethnic Disparities in Early Childhood Obesity." *Pediatrics* 141 (1): e20170865. https://doi.org/10.1542/peds.2017-0865.

Pan, Liping, Heidi M. Blanck, Sohyun Park, Deborah A. Galuska, Darcy S. Freedman, Ann Potter, Robert Petersen, and Flavia L. Trowbridge. 2016. "Trends in State/Territorial Obesity Prevalence by Race/Ethnicity Among Low-Income, Preschool-Aged Children in the United States, 2008–2011." *Pediatric Obesity* 11 (5): 397–402. https://doi.org/10.1111/ijpo.12084.